
PRiSM: Prototype Regularization for Few-Shot VLMs

Ghassen Baklouti*
ÉTS Montreal

Omprakash Chakraborty
ÉTS Montreal

Jose Dolz
ÉTS Montreal

Ismail Ben Ayed
ÉTS Montreal

Abstract

Training-free few-shot adaptation methods have gained significant attention recently in the context of Vision-language Models (VLMs). Yet, current benchmarks rely on strong assumptions about the statistics of the adaptation data, e.g., class balance. We question these simplifying assumptions and introduce a more realistic benchmark that varies both the levels of class balance and the effective number of classes in few-shot tasks via Dirichlet sampling. Surprisingly, under our setting, we observe substantial drops in the performances of state-of-the-art methods, more so when the number of labeled samples increases. To mitigate this, we introduce PRiSM, a class-prototype regularization that can be deployed as a plug and play module on top of any existing baseline method, significantly improving performances. Our method optimizes a novel multi-term loss, which includes a regularizer maximizing inter-class pairwise distances, along with additional terms promoting support-feature alignment and fidelity to the baseline prototypes. Furthermore, we introduce an effective and computationally efficient block Majorize-Minimize optimizer for our objective. More specifically, we derive a valid blockwise Lipschitz constant (i.e., a bound on the Hessian’s spectral norm), which can be computed efficiently via the Gershgorin circle theorem. Extensive experiments show that PRiSM improves several training-free baselines, with large gains when dealing with severe class imbalance and high numbers of classes.

1 Introduction

Vision–language models (VLMs) [20, 12, 25] have emerged as powerful general-purpose learners, enabling zero-shot recognition by aligning visual and textual representations in a shared embedding space. A common paradigm for downstream adaptation is few-shot learning, where a small set of labeled samples, referred to as the support set, is used to refine class prototypes, which are then evaluated on unseen query samples. Existing few-shot adaptation methods for VLMs can be broadly grouped into prompt-learning, adapter-based, and training-free approaches. Prompt-learning [29, 30] methods optimize learnable input tokens to obtain task-specific textual prototypes. In contrast, adapter-based methods [8, 26] learn a small set of trainable parameters to adjust the feature space, classifier weights, or prototype geometry while preserving the pretrained knowledge encoded in the model. Training-free approaches [1, 23, 28, 31] further improve efficiency by exploiting the support set without gradient updates, offering stable alternatives to training-based adaptation. Representative training-free methods include TIP-Adapter [28], which builds a cache model from support features; APE [31], which incorporates a feature-selection step into the cache model; GDA [23], which models each class as a Gaussian distribution with shared covariance in the embedding space; and ProKeR [1], which performs kernel-based prototype refinement.

Despite their strong empirical performance, these methods remain limited by two important factors. First, many training-free and adapter-based methods rely on sensitive task-specific hyperparameters that control the balance between the pretrained zero-shot classifier and the task-specific information provided by the support set [11, 21]. This trade-off is crucial: overemphasizing the support set

*ghassen.baklouti.1@ens.etsmtl.ca

can lead to overfitting and prototype drift, while relying too strongly on the zero-shot classifier can prevent meaningful task adaptation. Consequently, competitive performance often requires careful hyperparameter selection, typically through grid search on validation data, while hyperparameters tuned for one task rarely transfer reliably to another, requiring repeated calibration for each new dataset, shot number or adaptation scenario. This model-selection procedure is often unrealistic in genuine few-shot settings. A separate validation set may not be available, or is as expensive to annotate as the support set itself. In fact, assuming access to labeled validation sets may invalidate the few-shot assumption on which the methods are built. Indeed, in some cases, hyperparameters are selected using large validation or test sets [16, 28, 15], which implicitly assumes access to far more labeled data than the few-shot setting enables. Thus, when competitive performance depends on task-specific tuning, the reported results reflect not only the adaptation method itself, but also an additional model-selection protocol that departs from the few-shot assumption. Second, the standard few-shot evaluation protocol [1, 27, 29] relies on a critical yet often overlooked assumption: *class-balanced tasks*. In particular, both the support and query sets are constructed such that the classes are equally represented, implicitly enforcing a uniform prior. While this protocol simplifies adaptation, it poorly reflects real-world scenarios, where data distributions are naturally skewed, certain classes dominate others, and the set of observed classes during adaptation may be incomplete.

This discrepancy has important consequences. Under class-balanced sampling, the support set provides reliable and unbiased estimates of class prototypes, enabling effective adaptation even with very few samples. However, once this assumption is relaxed, the support distribution becomes highly uneven and prototype estimates are disproportionately influenced by the majority classes. As a result, the underlying representations become biased, leading to distorted decision boundaries and increased confusion among classes. Importantly, this issue is not necessarily mitigated by simply increasing the number of shots. On the contrary, as we observe in Fig. 1 (left), adding more samples under imbalanced distributions can further amplify the majority-class bias, causing several strong few-shot methods to degrade significantly. This behavior suggests that the core limitation of existing few-shot adaptation methods lies not in the adaptation mechanism itself, but in the *quality of the class prototypes* used for prediction. The increase in additional data with the shots may amplify the biases in prototype estimation rather than refine the decision boundaries.

To address these issues, we propose **PRiSM**, a validation-free *prototype regularization* framework that explicitly corrects biased and poorly separated class prototypes under realistic few-shot conditions. **PRiSM** is designed to mitigate the dominance of majority classes while preserving the discriminative structure of the VLM embedding space.

As illustrated in Fig. 1 (right), the impact of this regularization is very significant across both near balanced and severe imbalance settings. A key trend emerges from our results: baseline methods that substantially underperform under imbalance can reach performance levels comparable to stronger state-of-the-art approaches after prototype rectification. For instance, under severe imbalance, a relatively weaker method such as Tip-Adapter closes much of the gap to the recent ProKeR once its prototypes are regularized. This suggests that the primary bottleneck is not necessarily the adaptation mechanism itself, but the quality of the underlying prototypes. More broadly, this finding indicates that prototype bias induced by imbalance acts as a common failure mode across methods, and rectifying it can systematically elevate the performance of diverse approaches. Concretely, **PRiSM** regularizes the classifier prototypes by optimizing a novel loss that combines support-feature alignment, fidelity to the baseline prototypes and pairwise inter-class separation. We optimize our loss with an efficient block majorize–minimize algorithm, alternating between class prototype updates and a lightweight parameter-efficient finetuning of the vision encoder.

We summarize our contributions as follows:

- We expose a critical limitation of standard few-shot protocols for VLMs: class-balanced support and query sets may hide severe prototype bias. We introduce a controlled benchmark that varies both the effective number of classes and the level of class imbalance through Dirichlet sampling.
- We show that, under our realistic setting, increasing the number of shots degrades training-free adaptation as additional samples may amplify the majority-class bias.
- We introduce PRiSM, a class-prototype regularization that can be deployed as a plug-and-play module on top of any existing baseline method, significantly improving performances. Our method optimizes a novel multi-term loss, which includes a regularizer maximizing inter-

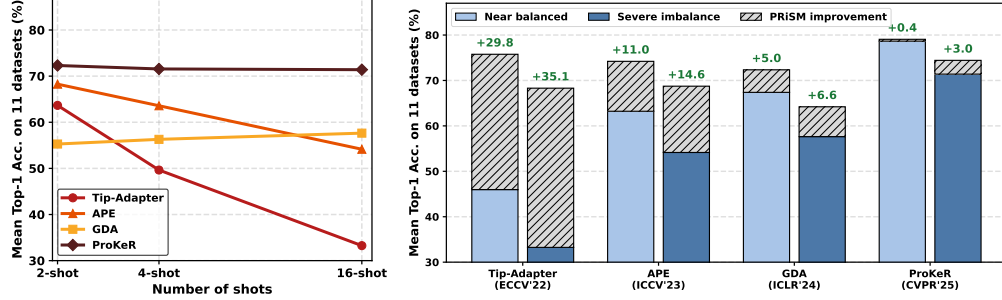


Figure 1: **Few-shot methods under realistic imbalance and effectiveness of prototype rectification.** (Left) Variation of mean Top-1 accuracy across 10 datasets with the number of shots under severe imbalance. Contrary to standard few-shot assumptions, increasing the number of support samples does not necessarily improve performance; instead, several strong baselines degrade as additional samples amplify majority-class bias in prototype estimation. (Right) Performance at 16-shot under near balanced and severe imbalance settings. Our proposed prototype rectification (**PRiSM**) consistently improves all baselines, with particularly large gains under severe imbalance.

class pairwise distances, along with additional terms promoting support-feature alignment and fidelity to the baseline prototypes.

- We introduce an effective and computationally efficient block Majorize-Minimize optimizer for our loss. More precisely, we derive a valid blockwise Lipschitz constant (i.e., a bound on the Hessian’s spectral norm), which can be evaluated effortlessly via the Gershgorin circle theorem.
- We report extensive experiments showing that **PRiSM** can improve the performance of few-shot baselines, including state-of-the-art methods, with large gains when dealing with severe class imbalance and high numbers of classes.

2 Toward Realistic Few-Shot Adaptation

Let us define a few-shot adaptation task from a downstream dataset with label space $\mathcal{Y} = \{1, 2, \dots, C\}$. Each task consists of a labeled support set $\mathcal{D}_{\text{adapt}} = \{(\mathbf{x}_n, \mathbf{y}_n)\}_{n=1}^N$, where $\mathbf{x}_n$ denotes an input image and $\mathbf{y}_n$ is its corresponding one-hot label vector over $\mathcal{Y}$. In the standard few-shot setting, $\mathcal{D}_{\text{adapt}}$ is constructed as a balanced support set with K labeled examples per class, where $K \in \{1, 2, 4, 8, 16\}$. Hence, the total number of adaptation samples is $N = KC$.

This balanced construction relies on several restrictive assumptions: all classes of interest are observed during adaptation, each class is represented by the same number of labeled examples, and the support and query sets follow the same class-frequency distribution. These assumptions are often violated in practical few-shot adaptation scenarios, where the available labeled data may cover only a subset of the target space and may exhibit class proportions that differ from those encountered at evaluation.

To capture these constraints, we consider realistic few-shot adaptation tasks in which the support set covers only an effective subset of classes $\mathcal{Y}_{\text{eff}} \subseteq \mathcal{Y}$, with $C_{\text{eff}} = |\mathcal{Y}_{\text{eff}}|$, and where the support and query sets may follow different class-frequency distributions. We define two class-coverage regimes according to the proportion of classes included in $\mathcal{Y}_{\text{eff}}$: *low* coverage, where 10%–30% of the classes are available during adaptation, and *high* coverage, where 70%–90% are available.

To model both class imbalance and distributional mismatch between adaptation and evaluation, we independently sample class proportions for the support and query sets from Dirichlet distributions over $\mathcal{Y}_{\text{eff}}$. Let $\boldsymbol{\pi}_S \sim \text{Dir}(\boldsymbol{\delta}_S)$ and $\boldsymbol{\pi}_Q \sim \text{Dir}(\boldsymbol{\delta}_Q)$ denote the class-proportion vectors used to construct $\mathcal{D}_{\text{adapt}}$ and $\mathcal{D}_{\text{query}}$, respectively. Since these proportions are sampled independently, the class frequencies observed during adaptation may differ from those encountered at evaluation.

The concentration parameter $\delta \in [0, 1]$ controls the dispersion of the sampled class proportions. Smaller values of δ produce highly skewed distributions dominated by a few classes, whereas larger values yield distributions closer to uniform sampling (see Appendix A.2). We consider two imbalance regimes: severe imbalance, with $\delta \in [0.1, 0.3]$, and near balanced sampling, with $\delta \approx 0.9$. For each

task, we fix the support set size to $N = K \times C_{\text{eff}}$ and the query set size to $N_{\text{query}} = 16 \times C_{\text{eff}}$ [3], and then allocate samples across classes according to π_S and π_Q . The resulting class-wise sample counts can vary substantially across both sets and some classes may even be absent in a query set for a given task. However, to ensure that the task remains a valid few-shot adaptation problem, we enforce that every class in $\mathcal{Y}_{\text{eff}}$ appears at least once in $\mathcal{D}_{\text{adapt}}$.

Combining the two class-coverage regimes with the two imbalance regimes yields **four** realistic evaluation settings: *low* coverage with *severe* imbalance, *low* coverage with *near balanced* sampling, *high* coverage with *severe* imbalance, and *high* coverage with *near balanced* sampling.

3 Formulation of PRISM

3.1 Our Method

Consider a pretrained vision-language model, such as CLIP, composed of an image encoder $f_\phi(\cdot)$ and a text encoder $f_\theta(\cdot)$, both mapping visual and textual inputs into a shared d -dimensional embedding space. Given a downstream task of image classification into C classes, zero-shot prediction amounts to matching image embeddings to text-based class prototypes. For each class c , let $\{\mathbf{t}_{m,c}\}_{m=1}^M$ denote a set of M textual descriptions (or prompts) of the class, e.g., “A photo of [class name]”. The corresponding zero-shot prototype is $\mathbf{u}_c = \frac{1}{M} \sum_m f_\theta(\mathbf{t}_{m,c}) \in \mathbb{R}^d$. Given a query image $\mathbf{x}$, with visual embedding $\mathbf{v} = f_\phi(\mathbf{x}) \in \mathbb{R}^d$, the zero-shot posterior probability prediction of class c is:

$$\Pr(c|\mathbf{x}) = \frac{\exp(\langle \mathbf{v}, \mathbf{u}_c \rangle / \tau)}{\sum_{i=1}^C \exp(\langle \mathbf{v}, \mathbf{u}_i \rangle / \tau)} \quad (1)$$

where pairwise operator $\langle \cdot, \cdot \rangle$ denotes the cosine similarity and τ is a temperature scaling parameter.

In general, few-shot adaptation methods update the visual features, the class prototypes or both using the support set (i.e., a set of few labeled samples), while discouraging substantial deviations from zero-shot class prototypes $\{\mathbf{u}_c\}_{c=1}^C$; see, for instance, [21, 28, 1]. We write such adaptation mechanisms in a generic form as $(\mathbf{a}, \mathbf{v}') = f_\psi(\mathbf{u}, \mathbf{v})$, where $\mathbf{u} = [\mathbf{u}_1, \dots, \mathbf{u}_C] \in \mathbb{R}^{dC}$ is a vector concatenating all zero-shot prototypes, $\mathbf{v}'$ is the adapted visual embedding, and ψ denotes the learnable or method-specific adaptation parameters. We denote by $\mathbf{a} = [\mathbf{a}_1, \dots, \mathbf{a}_C] \in \mathbb{R}^{dC}$ a vector concatenating the class prototypes obtained by a given baseline method. Thus, for each class c , $\mathbf{a}_c$ could be a task-adapted prototype obtained from any given few-shot adaptation method, or just the zero-shot prototype if no adaptation is used.

As discussed earlier, realistic few-shot settings often exhibit severe class imbalance, causing the support set to provide uneven evidence across the classes. Therefore, some initial prototypes might be less reliable than others: the minority-class prototypes might be weakly aligned with their visual samples, while the majority classes may dominate the classifier decisions, particularly when the support and query (test) distributions differ. To address this issue, our goal is to rectify a classifier $\mathbf{a}$ of a given baseline method by learning regularized prototypes $\mathbf{w} = [\mathbf{w}_1, \dots, \mathbf{w}_C] \in \mathbb{R}^{dC}$. For each class c , let $\boldsymbol{\mu}_c(\phi) = \frac{1}{n_c} \sum_n y_{n,c} f_\phi(\mathbf{x}_n)$ denote the mean of vision features within the class, where n_c is the number of class’ samples and $y_{n,c} \in \{0, 1\}$ is a binary-indicator function for the support-set samples: $y_{n,c} = 1$ when c is the class of the i^{th} support sample and $y_{n,c} = 0$ otherwise. We propose to minimize the following objective to learn rectified prototypes:

$$\mathcal{L}(\mathbf{w}, \phi) = \beta \sum_c \|\mathbf{w}_c - \boldsymbol{\mu}_c(\phi)\|_2^2 + \gamma \sum_c \|\mathbf{w}_c - \mathbf{a}_c\|_2^2 - \frac{\lambda}{2(C-1)} \sum_{c \neq c'} \|\mathbf{w}_c - \mathbf{w}_{c'}\|_2^2 \quad (2)$$

where $\|\cdot\|_2$ denotes the standard l_2 norm. The first term in Eq. (2) encourages alignment between each rectified prototype and the corresponding class’ visual mean, promoting consistency with the support-set distribution. The second anchors our prototype rectification to the baseline-method classifier, discouraging $\mathbf{w}_c$ from deviating substantially from $\mathbf{a}_c$. The last term is the pairwise regularizer driving our prototype rectification: It encourages inter-class separation by pushing away each pair of class prototypes, reducing their interference under our realistic settings. Note that this regularizer is

concave with respect to the class prototypes, which makes the overall objective non-convex² with respect to $\mathbf{w}$. Non-negative scalars β , γ , and λ control the relative strength of each term.

3.2 Block Majorize-Minimize optimization

Majorize-Minimize (MM) [14] is a general optimization paradigm that includes standard optimizers such as gradient descent, expectation-maximization and concave-convex procedures as special cases. At each iteration, MM minimizes a majorizing function – an upper bound on the original objective $\mathcal{L}(\mathbf{w})$ – which is tight at the current iteration j : $\mathcal{L}(\mathbf{w}) \leq \mathcal{M}(\mathbf{w}, \mathbf{w}^j)$ and $\mathcal{L}(\mathbf{w}^j) = \mathcal{M}(\mathbf{w}^j, \mathbf{w}^j)$. Hence, iterative step $\mathbf{w}^{j+1} = \min_{\mathbf{w}} \mathcal{M}(\mathbf{w}, \mathbf{w}^j)$ guarantees that the original objective does not increase: $\mathcal{L}(\mathbf{w}^{j+1}) \leq \mathcal{M}(\mathbf{w}^{j+1}, \mathbf{w}^j) \leq \mathcal{M}(\mathbf{w}^j, \mathbf{w}^j) = \mathcal{L}(\mathbf{w}^j)$. Hence, in the MM framework, step sizes are implicit, unlike standard gradient-descent updates, in which the learning rates are heavily searched using validation data and running the optimizer multiple times. In the following, we take advantage of the blockwise Lipschitz-gradient continuity of our objective in Eq. (2) with respect to the block of prototype variables $\mathbf{w}$. More precisely, we derive a bound on the spectral norm of the blockwise Hessian with respect to $\mathbf{w}$ (Prop. 1), which yields a block majorizing function with implicit step size. This relaxes the need for computationally intensive validation searches for the step size, while yielding results on par with the best learning rates obtained with validation data; See Fig. 2 in the experiments. Also, interestingly, our Lipschitz-driven step size is substantially larger than those commonly used in deep learning, yielding steeper descent towards a local minimum.

Our objective in Eq. 2 involves two blocks of variables: the corrected prototypes $\mathbf{w}$ and the vision encoder parameters ϕ . We optimize these variables using a block Majorize-Minimize (MM) procedure, which alternates two MM sub-steps: prototype rectification and feature adaptation. At each iteration, one block is updated while the other is kept fixed. This yields two simpler subproblems: a quadratic problem in $\mathbf{w}$ and a lightweight encoder-adaptation step in ϕ , with standard gradient steps³. The complete procedure is summarized in Algorithm 1 (refer Appendix A.5).

3.2.1 Optimization with respect to the prototypes

When vision-encoder variables ϕ are fixed, our objective in Eq. (2) reduces to a quadratic function of $\mathbf{w}$:

$$\mathcal{L}_w(\mathbf{w}) \stackrel{c}{=} \sum_{c,c'} \mathbf{R}_{cc'} \mathbf{w}_c^\top \mathbf{w}_{c'} - 2 \sum_c \mathbf{b}_c^\top \mathbf{w}_c = \mathbf{w}^\top \mathbf{\Phi} \mathbf{w} - 2 \mathbf{B}^\top \mathbf{w} \quad (3)$$

where symbol $\stackrel{c}{=}$ means equality up to a constant, $\mathbf{b}_c = \beta \boldsymbol{\mu}_c(\phi) + \gamma \mathbf{a}_c \in \mathbb{R}^d$, and $\mathbf{R} \in \mathbb{R}^{C \times C}$ is a prototype-interaction matrix given by $\mathbf{R}_{cc'} = \frac{\lambda}{C-1}$ for $c \neq c'$ and $\mathbf{R}_{cc} = \beta + \gamma - \lambda$. Also, matrix $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_C] \in \mathbb{R}^{dC \times dC}$, and matrix $\mathbf{\Phi} = \mathbf{R} \otimes \mathbf{I}_d$, where $\otimes$ denotes the Kronecker product and $\mathbf{I}_d$ is the $d \times d$ identity matrix.

As $\mathcal{L}_w(\mathbf{w})$ in Eq. (3) is quadratic, twice-differentiable function, it has a Lipschitz continuous gradient, i.e., there exists a strictly positive Lipschitz constant ρ such that the spectral norm (the largest singular value) of the Hessian is bounded by ρ : $\|\nabla^2 \mathcal{L}_w(\mathbf{w})\|_2 \leq \rho \forall \mathbf{w}$, with $\|\cdot\|_2$ denoting the spectral norm for a matrix. From this inequality, we obtain a majorizing function for $\mathcal{L}_w$ at iteration j :

$$\mathcal{L}_w(\mathbf{w}) \leq \mathcal{M}_w(\mathbf{w}, \mathbf{w}^j) = \mathcal{L}_w(\mathbf{w}^j) + \nabla \mathcal{L}_w(\mathbf{w}^j)^\top (\mathbf{w} - \mathbf{w}^j) + \frac{\rho}{2} \|\mathbf{w} - \mathbf{w}^j\|_2^2. \quad (4)$$

Minimizing the majorizing function $\mathcal{M}_w$ in Eq. (4) yields gradient-type updates with implicit step sizes $1/\rho$, which guarantee that the objective decreases by at least $\frac{1}{2\rho} \|\nabla \mathcal{L}(\mathbf{w}^j)\|^2$ at each iteration, a result known as the Descent Lemma. One valid Lipschitz constant would be the maximum singular value of the Hessian of Eq. (3), i.e. twice the spectral norm of $\mathbf{\Phi}$: $2\|\mathbf{\Phi}\|_2$. However, a spectral decomposition of $\mathbf{\Phi}$, a $dC \times dC$ matrix, could be computationally intensive for large numbers of classes. In the following proposition, we derive an explicit upper bound on $2\|\mathbf{\Phi}\|_2$, which provides a valid Lipschitz constant.

Proposition 1. *Objective $\mathcal{L}_w$ in Eq. (3) has ρ -Lipschitz gradient, with ρ given by:*

$$\rho = 2(|\beta + \gamma - \lambda| + \lambda) \geq 2\|\mathbf{\Phi}\|_2 \quad (5)$$

²In fact, the convexity of the overall objective in Eq. (2) is controlled by the choice of non-negative weighting factors β , γ and λ . However, choosing these factors such that the objective is strictly convex does not yield the desired prototype rectifications as it makes the l_2 -regularization components on $\mathbf{w}_c$ dominate the overall objective function.

³Standard gradient steps could be also viewed as an instance of MM updates.

Proof. The proof follows from the fact that $\|\Phi\|_2 = \|\mathbf{R} \otimes \mathbf{I}_d\|_2 = \|\mathbf{R}\|_2 \|\mathbf{I}_d\|_2 = \|\mathbf{R}\|_2$, and from bounding $\|\mathbf{R}\|_2$ using the Gershgorin’s circle theorem. The details are deferred to the appendix. $\square$

3.2.2 Optimization with respect to the vision encoder parameters

For the vision encoder block, with $\mathbf{w}$ fixed, the objective depends only on ϕ through the visual mean $\mu(\phi)$. We optimize this block by standard back-propagation using a conventional learning rate η_ϕ , while restricting the trainable parameters to a parameter-efficient subset of the image encoder. Specifically, we use LoRA [10] to introduce a small set of trainable low-rank updates in selected layers of $f_\phi(\cdot)$. For a frozen pretrained weight matrix $\mathbf{T}_0$, LoRA parameterizes the adapted weight as $\mathbf{T} = \mathbf{T}_0 + \mathbf{P}\mathbf{L}$ where $\mathbf{P} \in \mathbb{R}^{l_1 \times r}$ and $\mathbf{L} \in \mathbb{R}^{r \times l_2}$ are trainable low-rank factors with $r \ll \min(l_1, l_2)$. Only $\mathbf{P}$ and $\mathbf{L}$ are optimized, while $\mathbf{T}_0$ remains fixed. This parameter-efficient design reduces the number of trainable parameters while maintaining the expressive power of the pretrained vision encoder.

4 Experiments

Datasets. Following standard practices in vision-language models [20, 30, 29, 27], we consider 10 widely used image classification benchmarks covering diverse domains and levels of granularity. These include fine-grained recognition datasets such as Oxford-Pets [19], Stanford Cars [13], and FGVC Aircraft [17]; texture and material datasets including Describable Textures Dataset (DTD) [4] and Food-101 [2]; scene and remote sensing benchmarks such as SUN397 [24] and EuroSAT [9]; as well as generic object recognition datasets including Oxford Flowers [18], Caltech-101 [7], and UCF101 [22]. To avoid task-specific validation on the downstream benchmarks, we reserve ImageNet [5] exclusively as a held-out validation dataset for selecting a single hyperparameter configuration applied to all evaluation datasets.

Baselines. We evaluate **PRiSM** on top of four representative training-free few-shot adaptation methods for VLMs, covering diverse adaptation mechanisms. These include the cache-based retrieval method Tip-Adapter [28], Gaussian Discriminant Analysis (GDA) [23], Adaptive Prior refinement (APE), [31], and the recent kernel-based prototype refinement method ProKeR [1]. This diversity allows us to assess the generality of **PRiSM** as a plug-and-play prototype rectification module that can be applied across different adaptation paradigms.

Implementation Details. All experiments are conducted using CLIP [20] pre-trained features with a ViT-B/16 backbone [6]. For each downstream task, we first apply the corresponding baseline method using the task-specific support set, and then perform our prototype rectification on top of the resulting classifier. All reported results are averaged over 400 randomly sampled tasks. Data augmentation is applied during the feature extraction stage using random zoom, crops, and flips, following [1], with 10 augmentations per support sample. We use the same text prompts per dataset as in [1]. Following our claim that using a validation set for few-shot adaptation is unrealistic, we run all experiments using the same configuration across all datasets and shot numbers. Concretely, we set β to 0.01, γ to 1, and λ to 0.05. **PRiSM** is optimized for 3 alternating rounds; in each round, we perform 1 prototype-update step followed by 10 LoRA-update steps. We use LoRA adapters with rank 8 in the vision encoder, updating only the last three layers with a learning rate of 5×10^{-4} . All experiments are conducted on a single NVIDIA H100 SXM5 GPU.

4.1 Severe imbalance with high effective classes

Table 1 reveals a characteristic degradation of training-free few-shot adaptation baselines under realistic class imbalance. In particular, since C_{eff} is fixed to a large label space (80% of classes), increasing the number of shots introduces more support samples but does not alleviate the skewed Dirichlet distribution. As a result, majority classes become increasingly dominant, while minority classes remain under-represented, leading to progressively biased prototype estimates. This effect is clearly reflected in the performance degradation of several baselines as K increases. For instance, **Tip-Adapter drops from 63.66 to 33.25 when going from 2- to 16-shots**, while APE’s performance decreases from 68.26 to 54.15. In contrast, **PRiSM** explicitly rectifies these biased prototypes, leading

	Caltech	DTD	EuroSAT	FGVC	Food101	Flowers	Pets	Cars	Sun397	UCF101	Mean
16-shot											
Tip-Adapter	82.51	21.56	21.05	17.71	19.64	20.06	21.51	35.24	48.73	44.53	33.25
w/ PRiSM	93.24 _{+10.73}	61.76 _{+40.21}	74.36 _{+53.32}	31.40 _{+13.69}	33.92 _{+14.29}	84.98 _{+64.92}	91.04 _{+69.53}	68.10 _{+32.86}	68.20 _{+19.47}	76.03 _{+31.50}	68.30 _{+35.05}
GDA	87.68	44.33	58.34	23.08	55.80	63.59	66.29	50.72	60.31	66.26	57.64
w/ PRiSM	88.11 _{+0.43}	53.50 _{+9.17}	58.47 _{+0.12}	28.47 _{+5.39}	68.79 _{+12.99}	79.77 _{+16.18}	74.55 _{+8.26}	57.41 _{+6.69}	64.57 _{+4.25}	68.39 _{+2.13}	64.20 _{+6.56}
APE	91.67	50.77	34.10	19.26	60.34	41.50	69.25	44.17	62.08	68.38	54.15
w/ PRiSM	92.80 _{+1.13}	61.20 _{+10.44}	76.12 _{+42.02}	26.83 _{+7.57}	76.35 _{+16.01}	69.73 _{+28.23}	86.82 _{+17.57}	56.75 _{+12.58}	65.74 _{+3.66}	75.08 _{+6.71}	68.74 _{+14.59}
ProKeR	91.86	57.39	65.61	34.71	82.47	86.91	84.18	68.39	67.91	74.54	71.40
w/ PRiSM	92.80 _{+0.94}	62.75 _{+5.35}	80.65 _{+15.04}	35.13 _{+0.42}	81.51 _{-0.97}	88.94 _{+2.04}	90.10 _{+5.92}	68.38 _{-0.01}	67.50 _{-0.41}	76.51 _{+1.97}	74.43 _{+3.03}
4-shot											
Tip-Adapter	89.58	34.70	27.91	16.32	72.07	26.43	57.81	47.83	59.91	63.88	49.64
w/ PRiSM	92.97 _{+3.39}	59.89 _{+25.19}	78.12 _{+50.21}	31.00 _{+14.69}	82.33 _{+10.27}	84.24 _{+57.81}	90.41 _{+32.61}	67.12 _{+19.30}	67.26 _{+7.35}	75.27 _{+11.39}	72.86 _{+23.22}
GDA	86.12	45.04	58.32	23.23	54.52	68.29	62.83	45.71	54.45	64.43	56.30
w/ PRiSM	86.72 _{+0.60}	49.96 _{+4.92}	57.22 _{+1.11}	26.51 _{+3.27}	61.54 _{+7.03}	77.02 _{+8.73}	71.08 _{+8.25}	50.68 _{+4.97}	57.10 _{+2.65}	65.67 _{+1.23}	60.35 _{+4.05}
APE	92.11	49.91	52.32	19.19	84.13	58.52	86.95	55.07	65.92	71.68	63.58
w/ PRiSM	92.56 _{+0.45}	59.75 _{+9.84}	77.22 _{+24.90}	28.14 _{+8.94}	81.12 _{+3.01}	80.75 _{+22.23}	89.73 _{+2.78}	61.62 _{+6.55}	66.18 _{+0.26}	75.27 _{+3.59}	71.23 _{+7.65}
ProKeR	92.21	56.42	66.09	32.38	84.84	86.20	86.21	68.44	68.17	74.63	71.56
w/ PRiSM	92.62 _{+0.41}	60.53 _{+4.12}	79.01 _{+12.92}	33.42 _{+1.04}	81.16 _{+3.68}	87.64 _{+1.44}	88.93 _{+2.72}	66.49 _{-1.95}	66.15 _{-2.01}	75.21 _{+0.58}	73.12 _{+1.56}
2-shot											
Tip-Adapter	92.37	47.36	47.07	23.09	85.48	56.42	83.49	63.63	66.17	71.52	63.66
w/ PRiSM	92.92 _{+0.55}	58.53 _{+11.17}	76.37 _{+29.30}	30.61 _{+7.52}	81.88 _{+3.60}	83.88 _{+27.47}	89.69 _{+6.20}	66.36 _{+2.73}	66.27 _{+0.10}	75.37 _{+3.85}	72.19 _{+8.53}
GDA	84.41	43.48	57.08	23.15	57.06	70.03	58.75	43.46	52.70	62.69	55.28
w/ PRiSM	85.33 _{+0.93}	47.68 _{+4.20}	54.33 _{+2.75}	26.00 _{+2.85}	61.74 _{+4.68}	75.70 _{+5.67}	68.85 _{+10.10}	47.92 _{+4.46}	54.17 _{+1.47}	63.96 _{+1.27}	58.57 _{+3.29}
APE	93.05	49.03	59.16	25.65	87.01	73.49	90.30	65.95	67.44	71.52	68.26
w/ PRiSM	92.88 _{+0.17}	58.28 _{+9.25}	76.65 _{+17.49}	29.97 _{+4.31}	81.43 _{+5.59}	84.47 _{+10.98}	89.82 _{-0.48}	64.89 _{-1.07}	66.05 _{-1.39}	74.52 _{+3.00}	71.89 _{+3.63}
ProKeR	92.46	56.32	69.11	31.61	86.07	85.99	87.96	69.40	69.04	75.12	72.31
w/ PRiSM	92.49 _{+0.03}	58.55 _{+2.23}	77.78 _{+8.67}	31.36 _{-0.25}	80.82 _{-2.525}	86.91 _{+0.92}	88.54 _{+0.59}	65.60 _{-3.80}	65.22 _{-3.82}	74.92 _{-0.20}	72.22 _{-0.09}

Table 1: **Severe imbalance with high effective classes.** We report Top-1 accuracy (%) across 10 datasets. Differences with base methods are shown as subscripts.

to substantial recovery in performance under severe imbalance. For example, **PRiSM** improves the mean accuracy of Tip-Adapter by **+23.22** in 4-shot (**49.64** $\rightarrow$ **72.86**) and by **+35.05** in 16-shot (**33.25** $\rightarrow$ **68.30**), effectively reversing the degradation trend. Similarly, APE’s performance is enhanced by **+7.65** and **+14.59** in 4-shot and 16-shot, respectively, restoring performance despite the increasing imbalance. The effect is particularly pronounced on datasets where imbalance severely distorts the baseline prototypes. For instance, on EuroSAT and Flowers, **PRiSM** brings improvements to Tip-Adapter **ranging from nearly 50-70%** in 4- and 16-shot, whereas APE’s **performance gains range from 22.23% to 42.02%** under the same settings. Importantly, improvements are consistent across different adaptation mechanisms: GDA shows steady average gains of +3.29, +4.05, and +6.56 from 2-shot to 16-shot, while ProKeR, despite being a strong baseline, still benefits with gains of +1.56 and +3.03 in 4-shot and 16-shot scenarios. An interesting observation is that, simple methods such as Tip-Adapter, which substantially underperform arguably more complex approaches (i.e., ProKeR), are at par when **PRiSM** rectifies their visual prototypes (e.g., 72.19 vs. 72.31 in 2-shot; 72.86 vs. 71.56 in 4-shot), yielding state-of-the-art performance. These results show that **PRiSM** provides a simple yet effective plug-and-play correction that regularizes prototype geometry, preserving semantic fidelity while preventing collapse toward dominant classes.

4.2 Near-balanced setting with high effective classes.

In Table 2 we present the results under near-balanced class distributions, where the Dirichlet concentration is high (0.9) and class proportions are relatively uniform. In contrast to the severe imbalance regime, the degradation of baseline methods with increasing shots is less pronounced. This is expected, as the support set is more uniform across classes, leading to more stable prototype estimates. For instance, Tip-Adapter maintains relatively consistent performance from 68.80 (2-shot) to 62.98 (4-shot), before dropping to 45.98 at 16-shot, indicating that imbalance-induced bias is less but not entirely mitigated. Despite this improved stability, **PRiSM** consistently enhances performance across all methods and shot regimes. For Tip-Adapter, the mean accuracy improves substantially **from 62.98 to 74.90** in 4-shot (**+11.92**) and **from 45.98 to 75.77** in 16-shot (**+29.79**). Notably, even in this near-balanced setting, larger gains are observed on datasets such as EuroSAT (+34.56

	Caltech	DTD	EuroSAT	FGVC	Food101	Flowers	Pets	Cars	Sun397	UCF101	Mean
16-shot											
Tip-Adapter	86.43	33.16	33.32	21.19	53.77	34.60	37.75	44.95	56.69	57.91	45.98
w/ PRiSM	93.98 _{+7.54}	64.85 _{+31.69}	78.69 _{+45.37}	32.75 _{+11.56}	85.51 _{+31.74}	85.87 _{+51.27}	92.13 _{+54.38}	72.60 _{+27.65}	73.30 _{+16.60}	78.00 _{+20.09}	75.77 _{+29.79}
GDA	91.60	53.66	67.12	29.29	69.66	77.28	76.86	65.19	69.82	73.33	67.38
w/ PRiSM	91.85 _{+0.25}	62.52 _{+8.86}	65.98 _{-1.14}	34.49 _{+5.20}	78.42 _{+8.77}	89.53 _{+12.24}	83.47 _{+6.61}	69.20 _{+4.01}	72.81 _{+2.99}	75.15 _{+1.83}	72.34 _{+4.96}
APE	91.79	55.39	50.53	24.39	77.85	61.31	79.26	52.98	66.35	72.35	63.22
w/ PRiSM	93.58 _{+1.78}	65.61 _{+10.22}	83.42 _{+32.89}	31.14 _{+6.76}	83.85 _{+6.00}	79.72 _{+18.41}	90.91 _{+11.65}	64.26 _{+11.28}	71.66 _{+5.31}	77.96 _{+5.60}	74.21 _{+10.99}
ProKeR	93.84	67.33	81.40	43.17	86.36	93.34	89.56	77.45	73.70	80.39	78.65
w/ PRiSM	94.50 _{+0.66}	68.37 _{+1.04}	86.44 _{+5.04}	39.37 _{-3.80}	85.66 _{-0.71}	92.49 _{-0.85}	92.88 _{+3.32}	76.92 _{-0.53}	73.70 _{+0.00}	80.30 _{-0.09}	79.06 _{+0.41}
4-shot											
Tip-Adapter	91.50	49.39	48.13	23.01	83.82	55.90	79.50	60.75	66.37	71.46	62.98
w/ PRiSM	93.33 _{+1.83}	62.34 _{+12.95}	82.70 _{+34.56}	32.13 _{+9.12}	83.97 _{+0.14}	85.46 _{+29.57}	91.80 _{+12.30}	70.23 _{+9.49}	70.25 _{+3.87}	76.83 _{+5.37}	74.90 _{+11.92}
GDA	88.31	50.60	63.92	26.45	61.09	74.04	68.88	52.67	60.25	67.80	61.40
w/ PRiSM	88.49 _{+0.18}	55.34 _{+4.74}	62.20 _{-1.71}	29.94 _{+3.49}	68.39 _{+7.30}	83.04 _{+9.00}	76.83 _{+7.95}	56.89 _{+4.22}	62.56 _{+2.31}	69.03 _{+1.24}	65.27 _{+3.87}
APE	92.76	50.81	61.96	26.17	86.87	74.96	89.87	64.58	68.45	72.88	68.93
w/ PRiSM	93.38 _{+0.63}	62.72 _{+11.91}	81.46 _{+19.49}	32.43 _{+6.26}	83.93 _{-2.94}	86.23 _{+11.28}	91.44 _{+1.58}	68.26 _{+3.67}	70.02 _{-1.57}	77.08 _{+4.20}	74.69 _{+5.76}
ProKeR	93.10	61.67	76.62	36.12	86.24	89.75	88.65	72.70	71.18	77.57	75.36
w/ PRiSM	93.46 _{+0.36}	63.50 _{+1.82}	84.55 _{+7.93}	35.36 _{-0.76}	83.49 _{-2.76}	89.79 _{+0.04}	90.55 _{+1.89}	71.08 _{-1.62}	69.98 _{-1.20}	77.64 _{+0.07}	75.94 _{+0.58}
2-shot											
Tip-Adapter	92.67	50.55	59.23	28.02	87.06	72.99	89.03	68.38	67.77	72.31	68.80
w/ PRiSM	92.89 _{+0.21}	59.67 _{+9.12}	78.53 _{+19.30}	31.25 _{+3.23}	82.52 _{-4.54}	84.26 _{+11.27}	90.85 _{+1.82}	67.96 _{-0.42}	68.04 _{+0.28}	75.57 _{+3.26}	73.15 _{+4.35}
GDA	85.72	46.90	59.91	24.92	59.60	73.33	61.84	46.46	55.18	64.62	57.85
w/ PRiSM	86.41 _{+0.69}	51.32 _{+4.42}	56.89 _{-3.02}	27.77 _{+2.85}	64.26 _{+4.66}	78.49 _{+5.16}	71.76 _{+9.92}	50.94 _{+4.48}	56.57 _{+1.38}	65.80 _{+1.18}	61.02 _{+3.17}
APE	92.86	49.51	63.68	29.03	87.48	78.67	90.84	69.58	68.06	72.04	70.18
w/ PRiSM	92.80 _{-0.06}	60.20 _{+10.69}	78.44 _{+14.76}	31.22 _{+2.19}	82.38 _{-5.10}	85.71 _{+7.04}	90.36 _{-0.48}	67.52 _{-2.06}	67.73 _{-0.33}	75.84 _{+3.80}	73.22 _{+3.04}
ProKeR	92.89	58.87	73.60	33.71	86.59	87.70	89.58	71.38	70.38	76.84	74.15
w/ PRiSM	92.76 _{-0.13}	60.35 _{+1.48}	80.53 _{+6.93}	33.15 _{-0.56}	81.91 _{-4.68}	88.14 _{+0.44}	89.71 _{+0.13}	67.85 _{-3.53}	67.29 _{-3.09}	76.44 _{-0.40}	73.81 _{-0.34}

Table 2: **Near-balanced setting with high effective classes.** Top-1 accuracy (%) across 10 datasets.

in 4-shot, +45.37 in 16-shot) and Flowers (+29.57 in 4-shot, +51.27 in 16-shot), suggesting that prototype bias persists beyond extreme imbalance and can still significantly affect adaptation. For other baselines, the improvements are more moderate but remain consistent: GDA achieves gains of +4.96 for 16-shots, while APE improves by +10.99. Even for ProKeR, which already performs strongly, **PRiSM** provides slight gains in most cases (e.g., +0.58 in 4-shot and +0.41 in 16-shot), indicating that our method complements rather than replaces strong adaptation mechanisms.

4.3 Evaluation with Low effective classes.

We now report the average performance for low effective classes in Table 3. In this regime, the adaptation problem is intrinsically easier, as the support set covers a reduced label space, leading to more reliable class-wise statistics even under imbalance. Consequently, baseline methods exhibit significantly less degradation compared to the high effective class setting. Furthermore, higher performing approaches such as ProKeR remain relatively stable across shots (e.g., 85.47 → 83.74), indicating that the reduced label space reduces confusion among effective classes.

	Severe imbalance			Near balanced		
	2-shot	4-shot	16-shot	2-shot	4-shot	16-shot
Tip-Adapter	82.69	74.14	55.29	84.31	81.68	68.32
w/ PRiSM	86.14 _{+3.44}	86.52 _{+12.38}	87.03 _{+31.75}	86.71 _{+2.40}	88.02 _{+6.34}	88.81 _{+20.50}
GDA	74.21	75.93	77.40	76.07	79.79	83.82
w/ PRiSM	77.16 _{+2.95}	78.63 _{+2.69}	78.01 _{+0.61}	78.55 _{+2.48}	81.95 _{+2.16}	85.83 _{+2.01}
APE	83.91	82.91	79.44	84.42	84.71	83.37
w/ PRiSM	86.18 _{+2.27}	86.28 _{+3.36}	86.66 _{+7.22}	86.92 _{+2.50}	88.03 _{+3.31}	89.46 _{+6.09}
ProKeR	85.47	84.64	83.74	86.64	87.44	89.31
w/ PRiSM	85.91 _{+0.43}	86.30 _{+1.66}	87.32 _{+3.59}	86.73 _{+0.09}	88.38 _{+0.94}	90.49 _{+1.18}

Table 3: **Severe imbalance and near balanced settings with low effective classes.** We report mean Top-1 accuracy (%) across 10 datasets. Improvements over base methods are shown as subscripts; positive gains are highlighted in green.

Even in this easier regime, **PRiSM consistently improves performance across all methods and settings**. The gains are particularly pronounced for weaker baselines. For instance, Tip-Adapter benefits from substantial improvements ranging from **+12.38** (4-shot) to **+31.75** (16-shot) under severe imbalance, and from **+6.34** to **+20.50** under near-balanced conditions. This indicates that even when the number of effective classes is limited, imbalance still introduces biases in prototype estimation that accumulate with increasing shots. Even for ProKeR, which already produces well-structured prototypes, **PRiSM** still yields additional gains, particularly in higher-shot regimes (e.g., **+3.59** under severe imbalance at 16-shot). Overall, these results highlight a similar trend to the high effective class setting and portrays how **PRiSM** improves the performance across both severe and near-balanced regimes.

4.4 Varying Lipschitz learning rate.

In this section we analyze the impact of Lipschitz-based learning rate on performance. As discussed earlier, the proposed prototype rectification objective admits a quadratic form with a closed-form upper bound on its curvature. This allows us to derive a valid block Lipschitz constant ρ , which naturally induces a principled learning rate of $1/\rho$ for the prototype update. Unlike conventional hyperparameter tuning, this choice is directly tied to the geometry of the optimization landscape and guarantees stable MM updates. We conduct this analysis on top of ProKeR [1] under the 16-shot severe-imbalance setting with a high number of effective classes, using EuroSat, DTD and Flowers as representative datasets. To isolate the effect of the prototype-block step size, we vary only the prototype learning rate while keeping all other hyperparameters fixed. As observed from Fig. 2, performance exhibits a clear sensitivity to the learning rate. Extremely small learning rates under-optimize the prototype correction, leading to weaker alignment between the rectified prototypes and the support-set feature distribution. In contrast, the Lipschitz-based learning rate, indicated by the vertical dashed line, consistently matches the best-performing region across datasets. This indicates that the proposed step-size selection captures the effective curvature of the prototype rectification objective and provides a reliable operating point without requiring an exhaustive learning-rate sweep. Notably, the resulting step size is orders of magnitude larger than those typically used in deep learning, which often lie in the range $[10^{-4}, 10^{-2}]$; see [28] for instance.

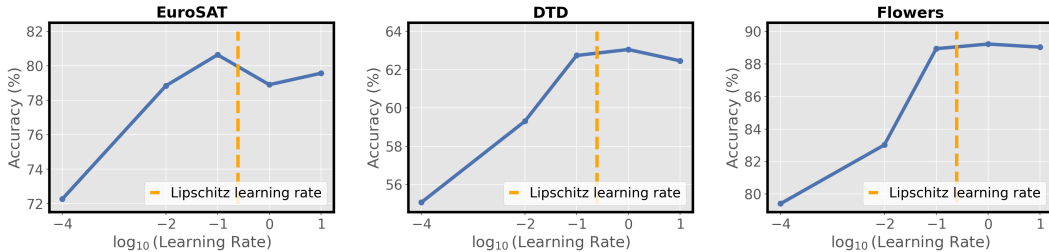


Figure 2: **Prototype-block performance as a function of different values of the learning rates.** The dotted vertical lines show the Lipschitz-based learning rate.

5 Conclusion

We revisited the evaluation of training-free few-shot adaptation for vision-language models by relaxing unrealistic assumptions on class balance and task composition. Through a Dirichlet-based benchmark, we showed that the performances of many state-of-the-art methods degrade significantly under class imbalance and large numbers of classes, even with increased supervision. To address this limitation, we propose **PRiSM**, a plug-and-play class-prototype regularization that consistently improves performance across diverse settings. **PRiSM** is optimized via an efficient block majorize-minimize algorithm with theoretically grounded Lipschitz-gradient bound, ensuring both effectiveness and scalability. Extensive experiments demonstrate that **PRiSM** substantially enhances robustness in challenging few-shot regimes. We hope that our benchmark and method will encourage more realistic evaluations and foster the development of adaptation techniques better suited to real-world deployment.

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A Supplementary

A.1 Proof of proposition 1

The spectral norm of the Hessian of quadratic objective $\mathcal{L}_w$ is given by:

$$2\|\Phi\|_2 = 2\|\mathbf{R} \otimes \mathbf{I}_d\|_2 = 2\|\mathbf{R}\|_2\|\mathbf{I}_d\|_2 = 2\|\mathbf{R}\|_2$$

The second equality above comes from the spectral-norm property of the Kronecker product, and the last comes from the fact $\|\mathbf{I}_d\|_2 = 1$.

Now we need to find an upper bound on $\|\mathbf{R}\|_2$. As $\mathbf{R}$ is symmetric, $\|\mathbf{R}\|_2$ is the largest absolute eigenvalue of $\mathbf{R}$. By Gershgorin's circle theorem, for each eigenvalue s of $\mathbf{R}$, there exists at least a row of the matrix such that s lies within a disk centered at the diagonal entry of the row, with a radius equal to the sum of absolute values of the off-diagonal elements in that row. Now observe that all the diagonal elements of $\mathbf{R}$ are given by

$$\mathbf{R}_{cc} = \beta + \gamma - \lambda \quad (6)$$

Also, for all rows of $\mathbf{R}$, the sum of absolute values of the non-diagonal elements is given by:

$$\sum_{c' \neq c} |\mathbf{R}_{c'c}| = (C-1) \frac{\lambda}{C-1} = \lambda \quad (7)$$

From (6) and (7), it follows that every eigenvalue s of $\mathbf{R}$ satisfies:

$$|s| \leq |\beta + \gamma - \lambda| + \lambda \quad (8)$$

Therefore:

$$\|\mathbf{R}\|_2 \leq |\beta + \gamma - \lambda| + \lambda \quad (9)$$

As $\|\Phi\|_2 = \|\mathbf{R}\|_2$, we obtain the following upper bound on the spectral norm of the Hessian of quadratic objective $\mathcal{L}_w$, which provides a valid Lipschitz constant:

$$\rho = 2(|\beta + \gamma - \lambda| + \lambda) \geq 2\|\mathbf{R}\|_2 \quad (10)$$

A.2 Effect of the Dirichlet Concentration Parameter on Class Imbalance

Figure 3 illustrates the effect of the Dirichlet concentration parameter δ on class imbalance across three classes. Smaller values of δ produce more skewed class distributions, whereas larger values lead to more uniform sampling.

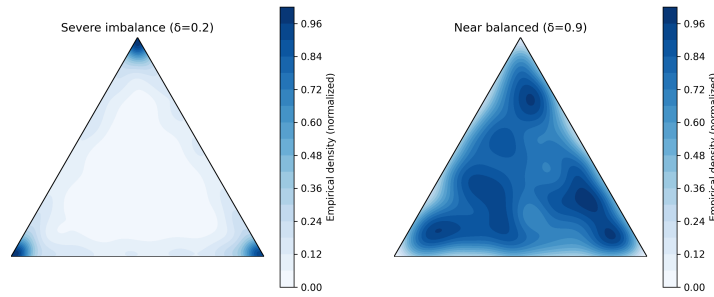


Figure 3: Effect of the Dirichlet concentration parameter δ on class imbalance across three different classes. Smaller δ values yield more skewed class distributions, while larger values approach uniform sampling.

A.3 Detailed Results for Near Balanced Setting with Low Effective Classes.

	Caltech	DTD	EuroSAT	FGVC	Food101	Flowers	Pets	Cars	Sun397	UCF101	Mean
16-shot											
Tip-Adapter	95.64	52.62	69.89	39.01	78.38	54.46	67.76	69.86	78.01	77.54	68.32
w/ PriSM	98.16 _{+2.52}	81.82 _{+29.20}	96.38 _{+26.48}	59.18 _{+20.17}	93.00 _{+14.62}	93.64 _{+39.18}	97.37 _{+29.61}	89.80 _{+19.94}	88.21 _{+10.19}	90.60 _{+13.06}	88.81 _{+20.50}
GDA	96.54	74.33	91.52	54.91	82.89	90.80	92.47	84.13	84.41	86.17	83.82
w/ PriSM	96.74 _{+0.20}	78.63 _{+4.29}	92.84 _{+1.32}	56.71 _{+1.81}	86.80 _{+3.91}	94.07 _{+3.27}	94.82 _{+2.35}	84.47 _{+0.34}	85.79 _{+1.38}	87.40 _{+1.24}	85.83 _{+2.01}
APE	97.80	72.76	87.23	48.52	93.47	84.48	95.54	81.81	85.33	86.72	83.37
w/ PriSM	98.25 _{+0.45}	82.64 _{+9.89}	97.31 _{+10.07}	61.06 _{+12.54}	93.67 _{+0.20}	96.32 _{+11.83}	97.75 _{+2.21}	88.75 _{+6.95}	88.23 _{+2.89}	90.61 _{+3.89}	89.46 _{+6.09}
ProKeR	98.16	81.49	94.37	63.41	93.79	97.02	96.03	90.51	88.01	90.28	89.31
w/ PriSM	98.43 _{+0.27}	84.44 _{+2.95}	96.99 _{+2.63}	64.98 _{+1.57}	93.67 _{-0.12}	97.49 _{+0.47}	97.48 _{+1.45}	91.54 _{+1.02}	88.72 _{+0.70}	91.18 _{+0.91}	90.49 _{+1.18}
4-shot											
Tip-Adapter	97.36	68.94	82.45	48.52	93.40	77.95	93.73	83.80	84.47	86.20	81.68
w/ PriSM	98.07 _{+0.71}	80.46 _{+11.52}	94.48 _{+12.03}	57.37 _{+8.85}	92.73 _{-0.66}	95.07 _{+17.11}	97.36 _{+3.64}	88.33 _{+4.53}	86.64 _{+2.17}	89.70 _{+3.50}	88.02 _{+6.34}
GDA	95.25	69.06	88.90	49.67	80.97	89.94	86.01	76.49	78.58	83.08	79.79
w/ PriSM	95.83 _{+0.57}	73.91 _{+4.86}	89.16 _{+0.26}	51.51 _{+1.84}	83.55 _{+2.58}	91.15 _{+1.21}	92.23 _{+6.22}	78.31 _{+1.81}	79.54 _{+0.96}	84.37 _{+1.29}	81.95 _{+2.16}
APE	97.99	69.38	89.63	53.13	94.52	87.71	96.99	86.88	85.07	85.85	84.71
w/ PriSM	98.13 _{+0.14}	80.29 _{+10.91}	94.67 _{+5.04}	57.42 _{+4.29}	92.31 _{-2.21}	95.89 _{+8.18}	97.10 _{+0.11}	87.80 _{+0.92}	86.71 _{+1.65}	89.93 _{+4.08}	88.03 _{+3.31}
ProKeR	97.87	77.66	92.12	57.14	93.82	95.29	96.76	87.91	86.65	89.17	87.44
w/ PriSM	98.11 _{+0.24}	81.37 _{+3.71}	94.97 _{+2.85}	58.87 _{+1.73}	92.26 _{-1.56}	96.75 _{+1.47}	97.27 _{+0.51}	88.08 _{+0.17}	86.19 _{+0.46}	89.92 _{+0.75}	88.38 _{+0.94}
2-shot											
Tip-Adapter	97.60	70.20	87.81	52.92	94.48	85.62	96.58	86.96	84.62	86.31	84.31
w/ PriSM	97.67 _{+0.08}	77.88 _{+7.68}	92.84 _{+5.02}	54.91 _{+1.99}	91.78 _{-2.70}	94.52 _{+8.89}	96.90 _{+0.32}	86.86 _{+0.10}	85.18 _{+0.56}	88.60 _{+2.29}	86.71 _{+2.40}
GDA	93.77	63.26	88.20	45.52	78.25	88.30	79.92	69.72	74.15	79.66	76.07
w/ PriSM	94.58 _{+0.81}	68.23 _{+4.97}	86.44 _{-1.76}	47.92 _{+2.40}	80.88 _{+2.63}	88.93 _{+0.63}	88.81 _{+8.89}	73.34 _{+3.62}	75.40 _{+1.24}	81.02 _{+1.36}	78.55 _{+2.48}
APE	97.90	67.59	88.01	54.63	94.68	86.30	97.25	87.64	84.62	85.63	84.42
w/ PriSM	97.91 _{+0.01}	78.05 _{+10.46}	93.77 _{+5.77}	56.21 _{+1.58}	91.70 _{-2.97}	94.77 _{+8.46}	96.74 _{-0.51}	86.42 _{-1.22}	85.22 _{+0.61}	88.42 _{+2.79}	86.92 _{+2.50}
ProKeR	97.73	75.37	89.66	56.53	93.85	94.18	96.34	87.66	86.21	88.84	86.64
w/ PriSM	97.84 _{+0.10}	77.88 _{+2.51}	91.86 _{+2.19}	56.52 _{+0.00}	91.38 _{-2.47}	95.49 _{+1.31}	96.36 _{+0.02}	86.37 _{-1.29}	84.66 _{-1.56}	88.91 _{+0.07}	86.73 _{+0.09}

Table 4: **Near-balanced setting with low effective classes.** We report Top-1 accuracy (%) across 10 datasets. Improvements over base methods are shown as subscripts; positive gains are highlighted in green.

A.4 Detailed Results for Severe Imbalance Setting with Low Effective Classes.

	Caltech	DTD	EuroSAT	FGVC	Food101	Flowers	Pets	Cars	Sun397	UCF101	Mean
16-shot											
Tip-Adapter	93.77	34.64	62.23	31.65	48.66	32.91	47.11	62.39	72.76	66.72	55.29
w/ PRiSM	98.15 _{+4.38}	79.01 _{+44.36}	93.55 _{+31.31}	55.29 _{+23.64}	92.08 _{+43.42}	93.81 _{+60.90}	96.55 _{+49.44}	87.28 _{+24.89}	85.61 _{+12.85}	88.98 _{+22.26}	87.03 _{+31.75}
GDA	95.38	65.95	86.62	47.51	74.50	84.59	87.17	73.98	77.05	81.26	77.40
w/ PRiSM	95.46 _{+0.08}	66.62 _{+0.67}	86.78 _{+0.16}	48.42 _{+0.90}	75.87 _{+1.37}	84.98 _{+0.39}	87.35 _{+0.18}	75.00 _{+1.02}	78.03 _{+0.99}	81.61 _{+0.36}	78.01 _{+0.61}
APE	97.85	69.35	79.98	41.27	90.60	74.27	93.48	78.44	83.82	85.31	79.44
w/ PRiSM	98.19 _{+0.34}	78.44 _{+9.09}	94.18 _{+14.20}	53.83 _{+12.56}	92.09 _{+1.49}	94.33 _{+20.06}	96.79 _{+3.30}	84.53 _{+6.09}	85.40 _{+1.58}	88.78 _{+3.47}	86.66 _{+7.22}
ProKeR	97.32	69.23	84.86	53.60	91.24	93.19	93.31	84.85	83.60	86.19	83.74
w/ PRiSM	97.89 _{+0.58}	79.35 _{+10.12}	93.39 _{+8.53}	58.12 _{+4.52}	92.03 _{+0.79}	95.92 _{+2.74}	96.77 _{+3.46}	86.45 _{+1.60}	84.35 _{+0.75}	88.98 _{+2.79}	87.32 _{+3.59}
4-shot											
Tip-Adapter	96.53	59.42	76.16	38.54	89.55	54.28	86.68	76.93	80.96	82.39	74.14
w/ PRiSM	97.74 _{+1.21}	78.14 _{+18.72}	92.33 _{+16.17}	54.49 _{+15.95}	91.50 _{+1.95}	93.96 _{+39.68}	96.94 _{+10.26}	86.64 _{+9.71}	84.84 _{+3.88}	88.65 _{+6.26}	86.52 _{+12.38}
GDA	93.63	62.28	87.78	45.50	77.19	86.02	82.54	70.23	73.77	80.41	75.93
w/ PRiSM	94.76 _{+1.12}	67.05 _{+4.77}	87.57 _{-0.21}	48.73 _{+3.23}	80.67 _{+3.48}	88.19 _{+2.17}	89.86 _{+7.32}	72.93 _{+2.70}	74.98 _{+1.21}	81.52 _{+1.11}	78.63 _{+2.69}
APE	97.89	68.16	87.09	48.46	94.16	82.97	96.56	84.70	83.95	85.22	82.91
w/ PRiSM	97.99 _{+0.10}	77.52 _{+9.36}	90.61 _{+3.52}	55.03 _{+6.58}	92.11 _{-2.05}	94.58 _{+11.61}	96.98 _{+0.42}	85.34 _{+0.65}	84.32 _{+0.37}	88.28 _{+3.06}	86.28 _{+3.36}
ProKeR	97.48	71.21	86.75	52.83	93.05	93.12	94.16	85.29	85.07	87.42	84.64
w/ PRiSM	97.77 _{+0.29}	77.40 _{+6.19}	90.49 _{+3.74}	56.19 _{+3.36}	91.59 _{-1.46}	95.33 _{+2.21}	95.93 _{+1.76}	85.32 _{+0.03}	84.22 _{-0.85}	88.81 _{+1.39}	86.30 _{+1.66}
2-shot											
Tip-Adapter	97.69	68.71	86.63	50.05	93.83	80.18	94.70	84.61	84.44	86.07	82.69
w/ PRiSM	97.80 _{+0.10}	77.08 _{-8.37}	91.99 _{+5.36}	54.46 _{+4.41}	91.38 _{-2.45}	94.25 _{+14.07}	96.00 _{+1.30}	85.77 _{+1.16}	84.44 _{-0.00}	88.19 _{+2.12}	86.14 _{+3.44}
GDA	92.69	60.92	86.95	43.99	77.02	86.46	78.12	66.25	72.07	77.61	74.21
w/ PRiSM	94.09 _{+1.40}	65.72 _{-4.80}	85.81 _{-1.14}	46.62 _{-2.63}	79.81 _{+2.79}	87.25 _{-0.79}	88.57 _{-10.44}	71.01 _{+4.76}	73.44 _{-1.38}	79.30 _{+1.69}	77.16 _{-2.95}
APE	97.84	67.39	86.62	53.32	94.58	84.63	97.00	87.49	84.67	85.55	83.91
w/ PRiSM	97.64 _{-0.20}	77.44 _{+10.05}	92.09 _{+5.48}	54.91 _{+1.59}	90.95 _{-3.63}	94.00 _{+9.37}	96.03 _{-0.97}	85.77 _{-1.72}	84.32 _{-0.35}	88.68 _{+3.13}	86.18 _{-2.27}
ProKeR	97.77	73.08	89.27	52.77	94.10	93.12	95.11	86.03	85.69	87.79	85.47
w/ PRiSM	97.72 _{-0.05}	76.64 _{+3.56}	92.69 _{+3.42}	54.13 _{+1.36}	91.56 _{-2.54}	94.77 _{+1.65}	95.43 _{+0.32}	84.56 _{-1.48}	83.77 _{-1.92}	87.80 _{+0.00}	85.91 _{+0.43}

Table 5: **Severe imbalance with low effective classes.** We report Top-1 accuracy (%) across 10 datasets. Improvements over base methods are shown as subscripts; positive gains are highlighted in green.

A.5 Block Coordinate Majorize-Minimize Algorithm

Algorithm 1: Block coordinate MM ($\mathbf{w}$, ϕ)

$\text{iter}_{\mathbf{w}} = 1$; $\text{iter}_{\phi} = 3$; $\beta = 0.01$; $\gamma = 1$; $\delta = 0.05$; $j = 3$

Initialize $\mathbf{w}^{0,0} = \mathbf{a}$ // Initial prototypes from the base method

Initialize $\phi^{0,0}$ // Pretrained vision encoder with LoRA parameters

for $j = 0, 1, \dots$ **do**

for $l_1 = 0, 1, \dots, \text{iter}_{\mathbf{w}}$ **do**

$\mathbf{w}^{j,l_1+1} = \mathbf{w}^{j,l_1} - \frac{1}{\rho} \nabla L_{\mathbf{w}}(\mathbf{w}^{j,l_1}, \phi^{j,0})$

 // $1/\rho$, from Eq. (5)

$\mathbf{w}^{j+1,0} = \mathbf{w}^{j,\text{iter}_{\mathbf{w}}}$

for $l_2 = 0, 1, \dots, \text{iter}_{\phi}$ **do**

$\phi^{j,l_2+1} = \phi^{j,l_2} - \eta_{\phi} \nabla L_{\phi}(\mathbf{w}^{j+1,0}, \phi^{j,l_2})$

 // η_{ϕ} , Standard learning rate for LoRA-based encoder adaptation

$\phi^{j+1,0} = \phi^{j,\text{iter}_{\phi}}$